

Experiment Day: Harmonic Oscillators

1 Starting point: spring and simple pendulum

install spring and simple pendulum

Discussion

- name similarities and differences of both systems
- What is the equilibrium position?
- What is the amplitude?
- What is one complete oscillation?
- Which system oscillates faster?
- What could affect the frequency?

Predictions - important quantities for frequency

- spring pendulum: mass m , spring constant k , amplitude A
- simple pendulum: mass m , string length l , amplitude A

Good Scientific Practice

- Change only **one parameter at a time**.
- Repeat every measurement several times.
- minimize error (e.g. measure 10 periods)
- Record your data immediately.
- Use the same measuring method throughout the experiment.
- Do not ignore measurements that do not fit your expectations. Instead, think about possible sources of error.

2 General Procedure

In science, experiments are performed to answer a specific question. To obtain reliable results, it is important to work carefully and systematically.

For every experiment, follow these steps:

1. State the research question.

What do you want to investigate?

2. **Make a prediction (hypothesis).**

What do you expect to happen? Explain your reasoning.

3. **Identify the variables.**

Decide which parameter you want to change (*independent variable*) and which quantity you will measure (*dependent variable*). Keep all other parameters constant to ensure a fair comparison.

4. **Perform the measurements.**

Measure the time for 10 oscillations instead of only one to reduce the influence of your reaction time. Repeat each measurement at least three times and calculate the average value.

5. **Calculate the period.**

$$T = \frac{t_{10}}{10}$$

6. **Calculate the frequency.**

$$f = \frac{1}{T}$$

7. **Analyze your results.**

Compare the measurements for the different parameter values. Look for trends and relationships.

8. **Draw a conclusion.**

Was your hypothesis correct? Which parameter influences the oscillation? Can you explain your observations?

3 Experiments

Experiment 1 – Spring Oscillator: Influence of the Mass

Question

- How does the mass affect the oscillation?

Materials

- spring
- stand with clamp
- different masses
- stopwatch

Procedure

- Keep the spring constant.
- Keep the amplitude approximately constant.
- Change only the mass.
- Measure the time for 10 oscillations.

Experiment 2 – Spring Oscillator: Influence of the Spring

Question

- How does the spring stiffness affect the oscillation?

Materials

- different springs
- one mass
- stand
- stopwatch

Procedure

- Keep the mass constant.
- Change only the spring.
- Measure the time for 10 oscillations.

Experiment 3 – Spring Oscillator: Influence of the Amplitude

Question

- Does the amplitude affect the oscillation?

Materials

- spring
- one mass
- stopwatch

Procedure

- Keep the spring and the mass constant.
- Change only the amplitude.
- Measure the time for 10 oscillations.

Experiment 4 – Simple Pendulum: Influence of the Length

Question

- How does the pendulum length affect the oscillation?

Materials

- pendulum bob
- string
- stand
- ruler

- stopwatch

Procedure

- Keep the mass constant.
- Change only the string length.
- Use small amplitudes.
- Measure the time for 10 oscillations.

Experiment 5 – Simple Pendulum: Influence of the Mass

Question

- Does the mass affect the oscillation?

Materials

- strings of equal length
- different masses
- stopwatch

Procedure

- Keep the string length constant.
- Keep the amplitude constant.
- Change only the mass.
- Measure the time for 10 oscillations.

Experiment 6 – Simple Pendulum: Influence of the Amplitude

Question

- Does the amplitude affect the oscillation?

Materials

- pendulum
- stopwatch

Procedure

- Keep the length and mass constant.
- Change only the amplitude.
- Measure the time for 10 oscillations.

4 Discussion

Discuss the following questions:

- Which parameters influence the frequency?
- Which parameters have no influence?
- Which results surprised you?
- How are the spring oscillator and the simple pendulum similar?
- How do they differ?

5 Similarities and Differences

Similarities

Both the spring oscillator and the simple pendulum share many common properties:

- Both are **harmonic oscillators** (for small oscillations).
- Both oscillate around an **equilibrium position**.
- Both are characterized by an **amplitude**, a **period**, and a **frequency**.
- Both can be described using **sine and cosine functions**.
- In both systems, the **restoring force** always points back towards the equilibrium position.
- Because of **inertia**, the object overshoots the equilibrium position, leading to continuous oscillations.

Differences

Although both systems behave similarly, the restoring force and the parameters determining the frequency are different.

Restoring force

- **Spring oscillator:** The restoring force is provided by the spring and is described by Hooke's law, $F = -kx$.
- **Simple pendulum:** The restoring force is provided by gravity.

Frequency The frequency of a spring oscillator is $f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$, where k is the spring constant and m is the mass.

The frequency of a simple pendulum is $f = \frac{1}{2\pi} \sqrt{\frac{g}{l}}$, where g is the gravitational acceleration and l is the pendulum length.

Comparison

- For the **spring oscillator**, the frequency depends on the **mass** and the **spring constant**.
- For the **simple pendulum**, the frequency depends on the **pendulum length** and the **gravitational acceleration**.
- Increasing the mass decreases the frequency of a spring oscillator, whereas the mass has **no influence** on the frequency of a simple pendulum.
- A stiffer spring increases the frequency, while a longer pendulum decreases the frequency.

Key Takeaway Although the spring oscillator and the simple pendulum look quite different, they follow the same mathematical model of harmonic motion. The main difference is which physical quantities determine their frequency.